

Chain and Independent Food Service Business Performance According to Income and Neighborhood in Boston



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Data Science Major Capstone

Background and Aims

- Food health and safety are essential so that illness does not spread.
- Investigating differences in food handling practices across neighborhoods helps better understand whether there are trends between socioeconomic class, neighborhood, and food safety.
- Previous study in Philadelphia and British Columbia, Canada show that chain restaurants have fewer total violations per inspection than non-chain restaurants (Tung, 2018). Non-chain restaurants generally have a higher proportion of people living in poverty, and residents live in lower income with more Black and Hispanic residents near them (Leinwand et al., 2017).
- This study is interested in the relationship between chain food inspection violations and the neighborhoods in which they reside.

Research Question

To what extent do chains perform differently (in terms of violations) than independent businesses in different neighborhoods based on income in Boston in 2024 to 2026?

Data Summary

Table 1: Data characteristics and cleaning

Stage	Rows	Explanation
1. Original dataset	433,894	Original data, one row per violation.
2. Spatial join with census tract	406,553	Map coordinates of businesses to census tract polygons. Some businesses had null coordinates.
3. Income merge	406,553	Median income in each businesses' census tract added to each row. 13 tracts had no ACS income data. 9 are special purpose tracts and 4 have ACS reporting gaps. Rows without data have NaN for income.
4. 2024-2026 filter	78,242	Rows dropped either outside time range or null.
5. Deduplication	17,080	Deduplicate to have one row per inspection visit. A new column gives the number of violations per visit.

Results and Discussion

- There are fewer violations in high income neighborhoods than there are in low-income neighborhoods (Figure 2).
- Negative binomial mixed effects model was used due to overdispersion in the Poisson.
- For brand chains in the lowest income neighborhood, there are 5.01 predicted violations. For brand chains in the highest income neighborhood, there are 2.94 predicted violations.
- Brand chains have 23% more violations than independents in low-income neighborhoods. Brand chains have the most variability based on neighborhood income.
- There are 4.08 predicted violations for an independent businesses compared to the average of 4.45 violations.
- National chains have 31% fewer violations than Independents across all neighborhood incomes, suggesting efficacy of corporate policies

Limitations

- There is potentially temporal and spatial correlation in the model. Two tracts next to each other probably have more similarities than another on the other side of the city. The model does not account for trends over time.
- Independents comprise the majority of inspection visits. It is much more difficult to make accurate predictions for chains because there is much less data than independents (Figure 1).
- A few chains dominate the National Chain category, meaning that these few chains' outcomes represent the entire group.

References

Data Source: Food Establishment Inspections. (n.d.). [Tabular]. Inspectional Services Department. Retrieved <https://data.boston.gov/dataset/food-establishment-inspections>

1. Leinwand, S. E., Glanz, K., Keenan, B. T., & Branas, C. C. (2017). Inspection Frequency, Sociodemographic Factors, and Food Safety Violations in Chain and Nonchain Restaurants, Philadelphia, Pennsylvania, 2013-2014. *Public Health Reports*, 132(2), 180-187. <https://doi.org/10.1177/0033354916687741>

2. Tung, J., BCIT School of Health Sciences, Environmental Health, & Heacock, H. (2018). Critical food safety violations in Surrey: relationship to community median household income and restaurant type. *BCIT Environmental Public Health Journal*. <https://doi.org/10.47339/ephj.2018.57>

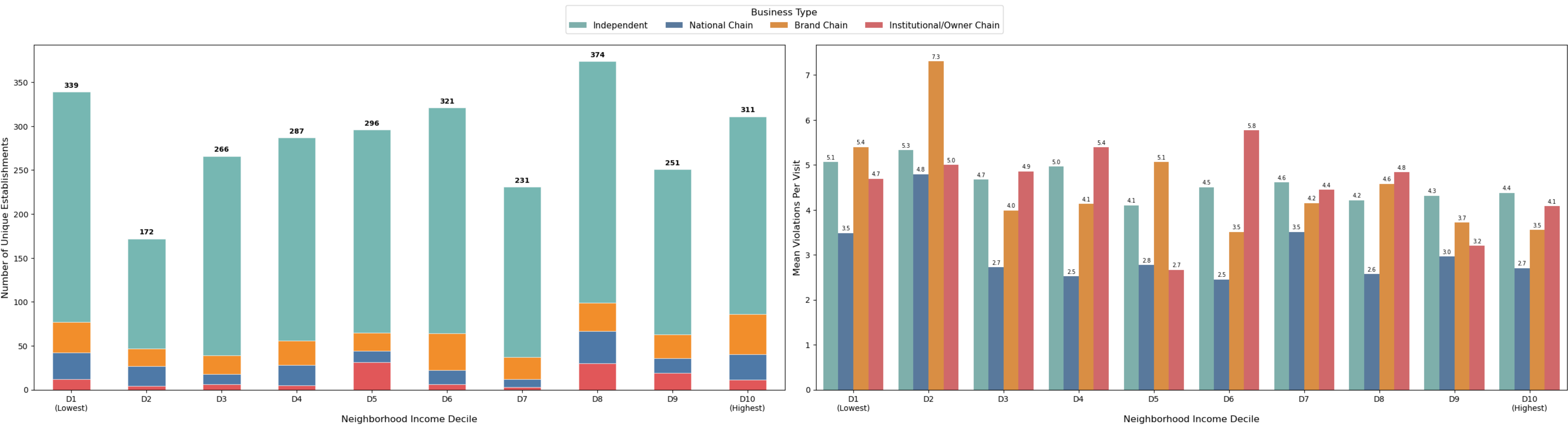


Figure 1: Unique Food Establishments by Business Type and Neighborhood Income Decile

Figure 2: Mean Violations Per Visit by Business Type and Neighborhood Income Decile